

TL;DR

MSCS2202 Machine Learning Kaggle Regression Challenge

Main idea: predict a continuous target value using 20 numerical features (x0-x19). The best result came from a simple, interpretable pipeline: mean imputation for missing values + Linear Regression.

0.507384

Local validation R^2

0.57123

Public R^2

0.54056

Private R^2



Official final rank: #3 on the private leaderboard

Ebunoluwa Adejumo

EDA: dataset structure

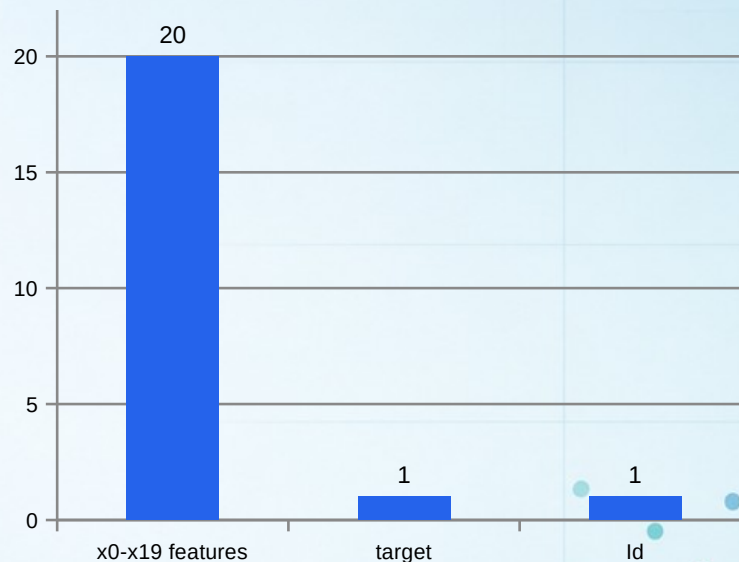
Initial inspection focused on dimensions, variables, target availability, and the Kaggle submission format.

Training data

2,500 rows
22 columns
20 features + target + Id

Test data

2,500 rows
21 columns
20 features + Id
No target column



Problem type

Supervised regression

The model learns from rows where target is known, then predicts target for the test rows where Kaggle hides the answer.

Submission format

Id,target
0,prediction
1,prediction
2,prediction

EDA: missing values and preprocessing

Missing data was the main preprocessing issue in the provided dataset.

491

Missing values in training features

509

Missing values in test features

0

Missing values in target

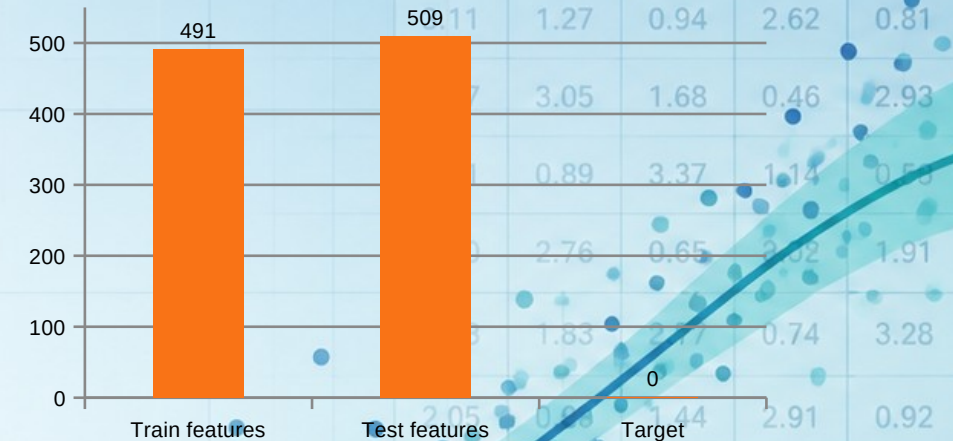
Preprocessing motivation

Scikit-Learn Linear Regression cannot train directly on NaN values. Therefore, missing features had to be filled before model fitting.

Validation design

1. Split train data into 80% train / 20% validation
2. Fit imputer only on the training split
3. Transform validation using the training imputation values
4. Compare models using R^2

Mean imputation selected for final model



Experiments and motivation

Each experiment tested a specific way to improve the regression score.

Approach

Median imputation + Linear Regression
Mean imputation + Linear Regression
Ridge Regression
Lasso Regression
Elastic Net
Polynomial Ridge

Motivation

Create a simple baseline
Compare missing-value strategies
Test L2 regularization
Test L1 regularization / coefficient shrinkage
Combine Ridge and Lasso penalties
Test nonlinear interactions and squared terms

Model selection principle

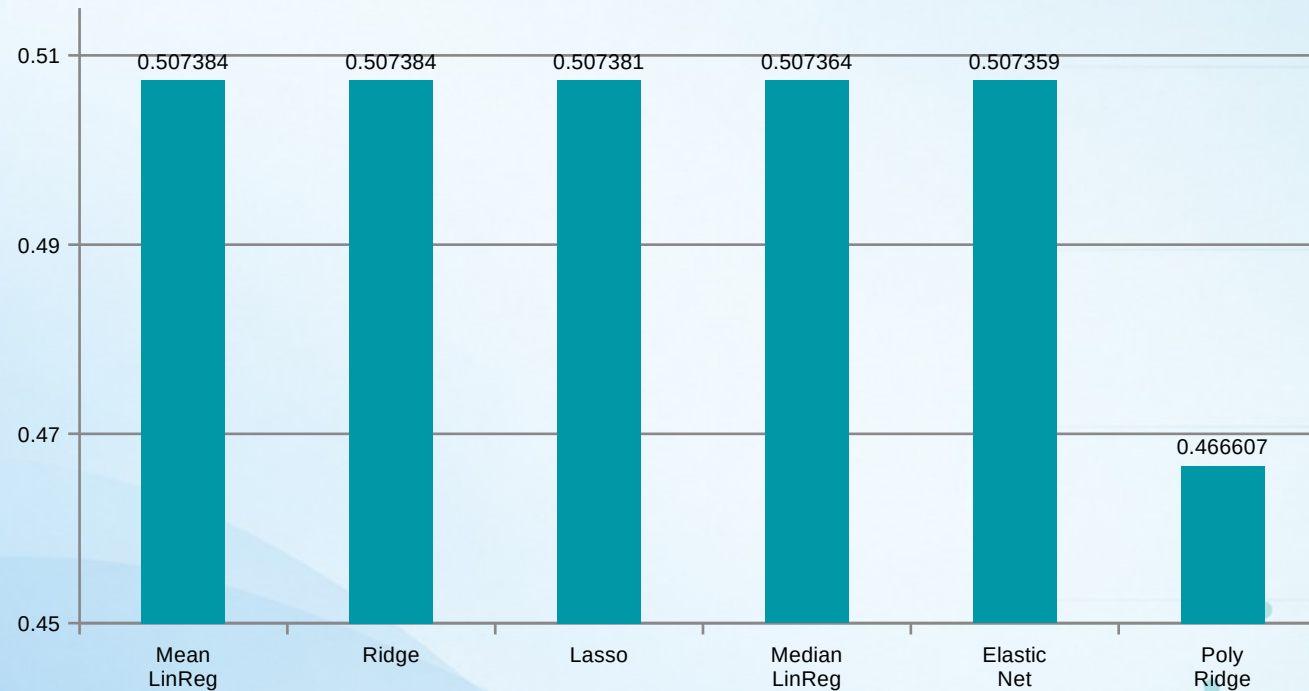
Use the same validation split and R^2 metric to compare models fairly. Select the approach that performs best on validation and Kaggle results.

Result insight

Additional complexity did not improve the score. The strongest final model was still simple and interpretable.

Model comparison: validation R^2

Validation results showed that Linear Regression with mean imputation was the best local approach.



Best validation scores

Model	R^2
Mean Imputation + LinReg	0.507384
Ridge Regression	0.507384
Lasso Regression	0.507381
Median Imputation + LinReg	0.507364
Elastic Net	0.507359
Polynomial Ridge	0.466607

Interpretation: Ridge, Lasso, and Elastic Net were close to ordinary Linear Regression, but they did not meaningfully improve the validation R^2 . Polynomial features reduced performance.

Leaderboard performance

The final competition standing is based on the private leaderboard.



0.57123

Public R^2

#6

Public rank

Kaggle split

Public leaderboard:
approximately 48% of the
test data

Private leaderboard:
approximately 52% of the
test data

Private leaderboard reflects
the final standings after
completion.

0.54056

Private R^2

#3

Final rank

Kaggle submissions

1. Median imputation + LinReg: public R^2 = 0.57121
2. Mean imputation + LinReg: public R^2 = 0.57123

Final approach and code link

The code used to generate the final score is available in the GitHub notebook below.

Final pipeline

1. Load train and test datasets
2. Separate x0-x19 features from target
3. Split training data for validation
4. Fit mean imputer on training data
5. Train Linear Regression
6. Retrain on full training data
7. Predict test targets and submit Id,target CSV

Final private leaderboard rank: #3

Selected model

Linear Regression + Mean Imputation

Why this final model? It had the strongest validation result, produced the best public Kaggle score among my submissions, and remained simple enough to explain clearly.

GitHub code link

[Machine Learning Regression Challenge.ipynb](https://github.com/Ebun25/MSCS2202_Machine_Learning_Kaggle_Challenge/blob/main/Machine_Learning_Regression_Challenge.ipynb)

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